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ICCUB

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UNIVERSITAT DE BARCELONA

Gauge Theories for the Standard Model

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Relativity + Quantum Mechanics + Causality + Cluster Decomposition

= **Quantum Field Theory**

Lorentz-Invariant S-matrix + Unitarity + massless particles

= **Gauge Theory** (for spin 1)

= **General Relativity** (for spin 2)

(Weinberg 1964)

It seems we do not have a choice!!

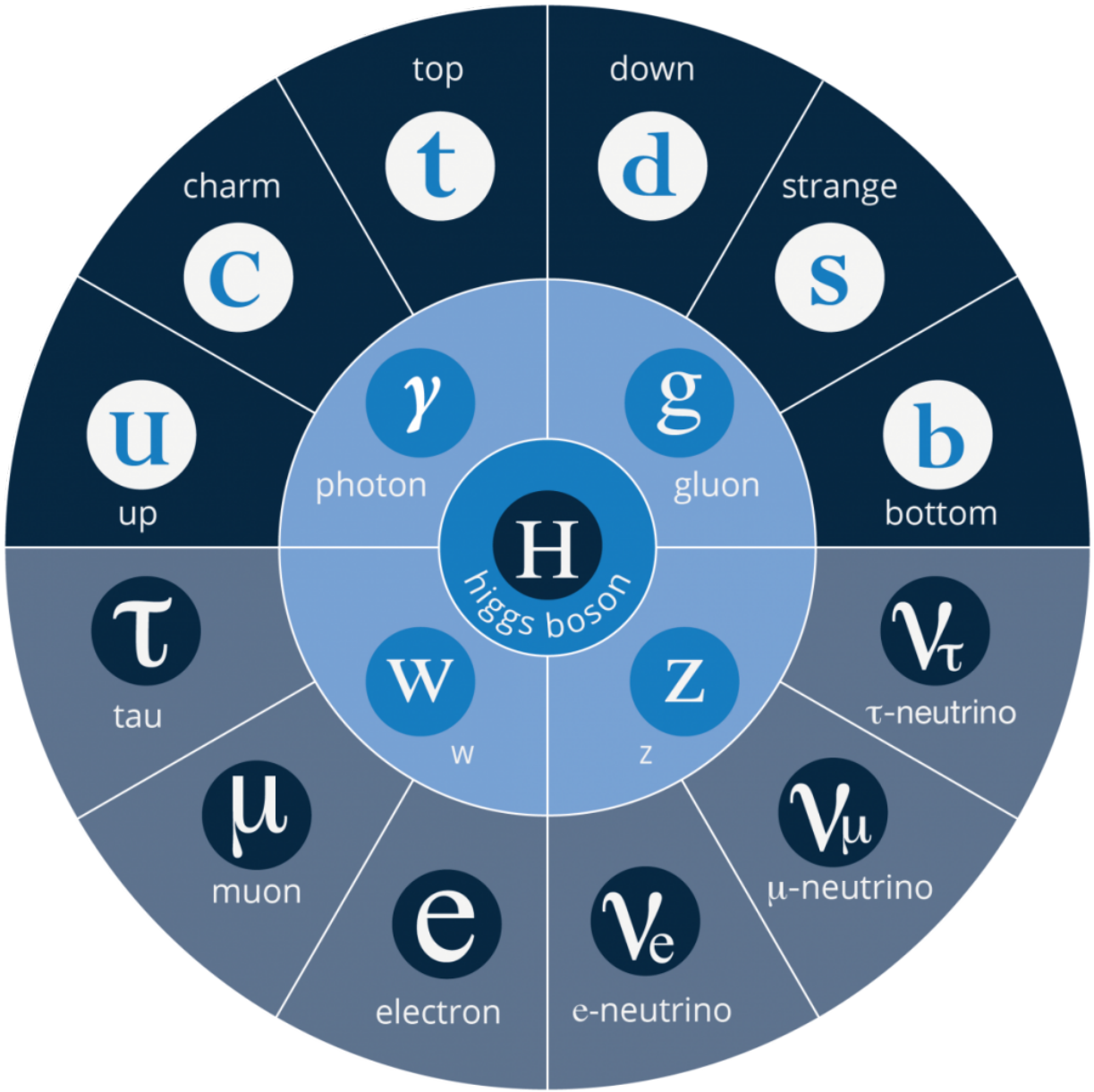
The above + Observed particle content

= Standard Model of Particle Physics:

A **Renormalizable Gauge Quantum Field Theory** with the following ingredients:

1. Gauge group $\mathcal{G}_{\text{SM}} \equiv SU(3)_c \times SU(2)_L \times U(1)_Y$.
2. Spontaneus symmetry breaking (SSB) of $\mathcal{G}_{\text{SM}}$ down to $SU(3)_c \times U(1)_{\text{em}}$ via the vacuum spectation value of a single $SU(2)_L$ doublet.
3. Three families of Matter fermions in specific representations of $\mathcal{G}_{\text{SM}}$.

	Q	u	d	L	e	H
$SU(3)_c$	$\mathbf{3}$	$\mathbf{3}$	$\mathbf{3}$	$\mathbf{1}$	$\mathbf{1}$	$\mathbf{1}$
$SU(2)_L$	$\mathbf{2}$	$\mathbf{1}$	$\mathbf{1}$	$\mathbf{2}$	$\mathbf{1}$	$\mathbf{2}$
$U(1)_Y$	$1/6$	$2/3$	$-1/3$	$-1/2$	-1	$1/2$


top
t
down
d
charm
c
strange
s
u
up
photon
γ
gluon
g
b
bottom
H
higgs boson
τ
tau
W
w
Z
z
τ-neutrino
μ
muon
μ-neutrino
electron
e
e-neutrino

OBJECTIVE

The purpose of this course is

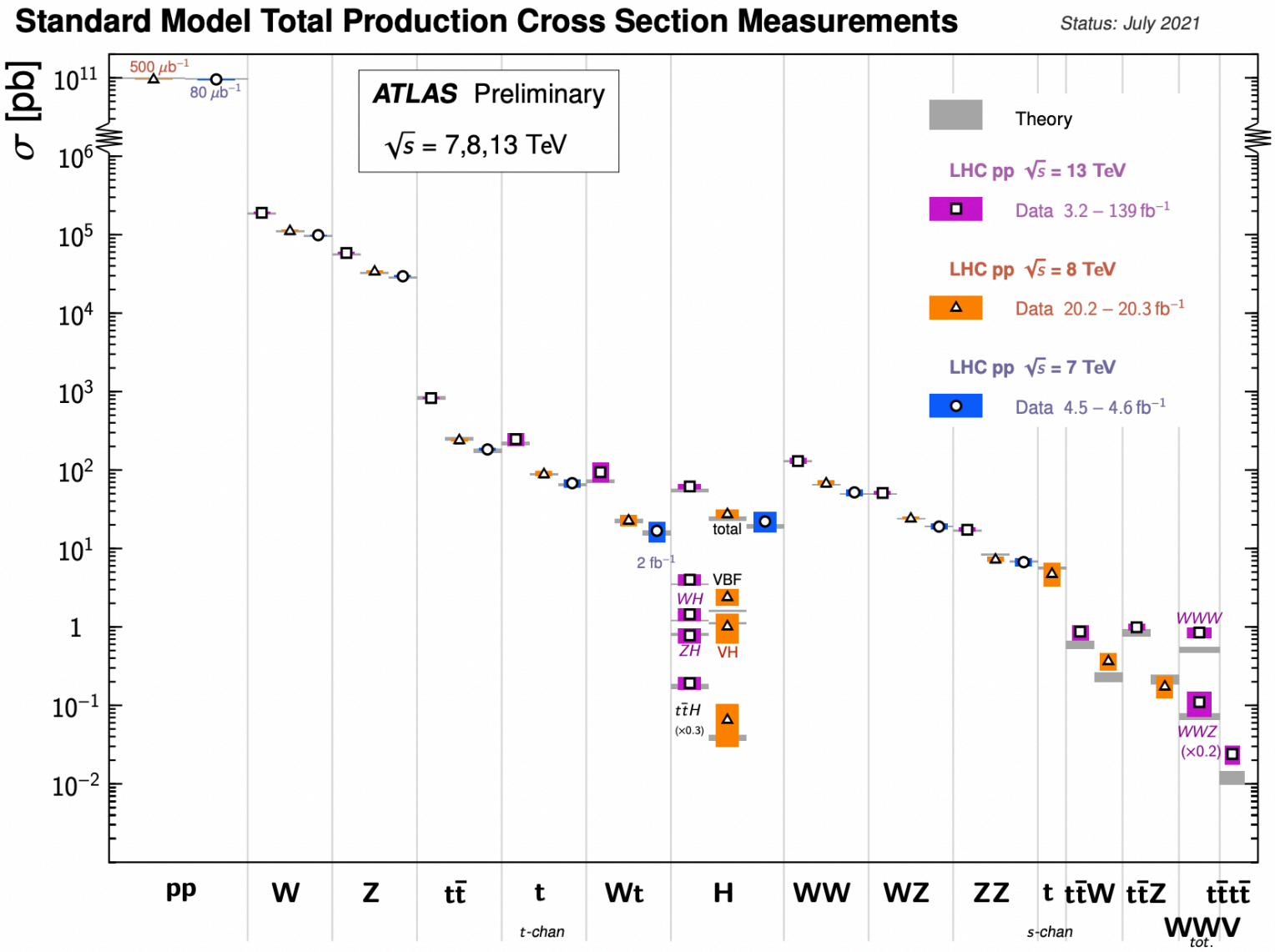
To understand the SM in detail at the **quantum** level

Of course, this will imply understanding a wider (quite general) set of gauge theories with

- different gauge groups
- different patterns of SSB, and
- different field content

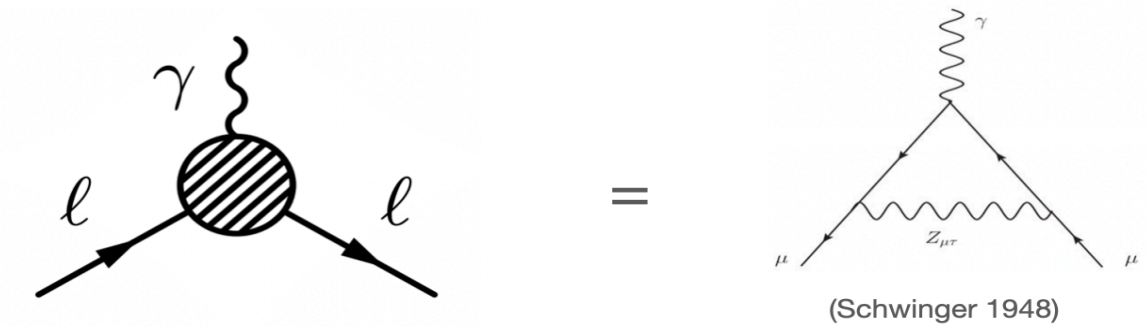
Thus we will also be able to understand e.g. **Grand Unified Theories** such as Georgi-Glashow's $SU(5)$, and much more!

The Standard Model is **A-MA-ZING**

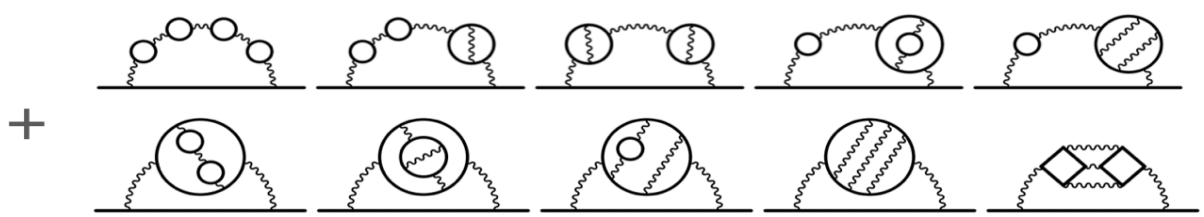


The Standard Model is **A-MA-ZING**

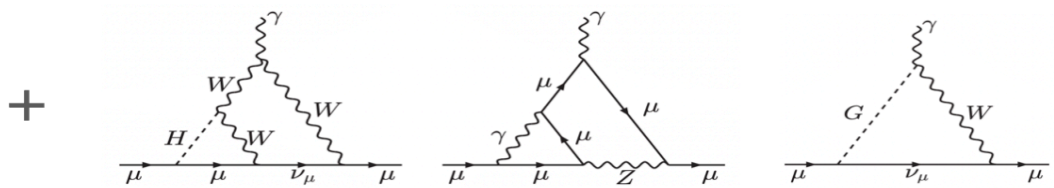
- Muon Magetic Moment:



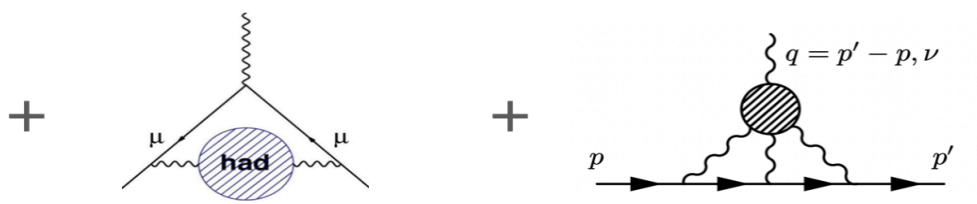
(Schwinger 1948)



(QED up to 5 loops)



(Electroweak and Higgs)



(Hadronic VP and LbL (QCD))

The Standard Model is **A-MA-ZING**

- Muon Magetic Moment: (known to parts-per-billion)

$a_\mu = (g_\mu - 2)/2$	
Exp	0.00116592089
Schwinger 1948	$\alpha/2\pi = 0.00116140000$
QED (5 loops)	0.00116584719
Electroweak	0.00000000154
Hadronic VP	0.00000006845
Hadronic LbL	0.00000000092
Total SM	0.00116591810

Current status: $a_\mu^{\text{exp}} - a_\mu^{\text{SM}} = 0.00000000279(76)$ BSM???

STRUCTURE OF THE COURSE

PART I: 25 hrs (Prof. Javier Virto)

Gauge Theories with and without SSB, and their quantization

PART II: 13 hrs (Prof. Domènec Espriu)

Renormalization and Quantum Chromodynamics

PART III: 13 hrs (Prof. Miguel Angel Escobedo)

Electroweak Theory and Higgs

STRUCTURE OF THE COURSE

PART I: 25 hrs (Prof. Javier Virto)

Gauge Theories with and without SSB, and their quantization

1. Massless spin-1 particles and Gauge Symmetry
2. Non-Abelian Gauge Theory
3. Quantization of Yang-Mills Theory
4. Gauge Theories with Spontaneous Symmetry Breaking
5. QFT Anomalies

PART II: 13 hrs (Prof. Domènec Espriu)

Renormalization and Quantum Chromodynamics

6. Radiative corrections in Gauge Theory
7. The limits of perturbation theory

PART III: 13 hrs (Prof. Miguel Angel Escobedo)

Electroweak Theory and Higgs

8. Gauge structure of Electroweak Theory
9. The Electroweak Theory beyond tree level
10. Radiative corrections in Electroweak Theory

YOU WILL LEARN ...

- Why we need Gauge Symmetry
- Why some Gauge Theories have *ghost* particles ...
- What happens when the quantum vacuum is not gauge invariant
- Why quantum mechanics breaks some global symmetries ...
- ... and why it does **not** break the Standard Model gauge symmetry !!
- Why QCD is asymptotically free (explicitly!)
- Why Flavor-changing processes are loop-mediated, and the GIM mechanism
- Why pions are Goldstone bosons, and why $\pi^0 \to 2\gamma$ occurs
- AND MUCH MORE...

EVALUATION

For the **evaluation** there will be several problem *hand-outs* during the course.

The final grade will be based on these hand-outs, in the proportion

$$\text{Grade} = 50\% \text{ (Part I)} + 25\% \text{ (Part II)} + 25\% \text{ (Part III)}$$

There will be **no final exam**.

INSTRUCTORS



Domènec Espriu Climent

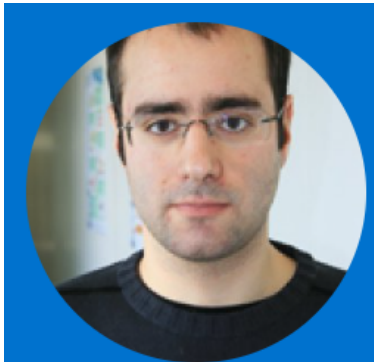
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